AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 36 · STUDENT LESSON

Research defense and next questions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Defend what was learned and the next question the evidence justifies

Separate descriptive results, computational consistency and untested generalization while proposing a controlled follow-up.

Keep one habit: Question and contract → complete main ledger → separate picture → boundary probes → investigation → independent attempt.

MAIN LESSON · SECTION 1

Defend what was learned and the next question the evidence justifies

> Separate descriptive results, computational consistency and untested generalization while proposing a controlled follow-up.

Learning goals

I can explain the **mechanism** and terminology of **research defense** and **next question**s.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Presentation and challenge session. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Independent reproduction check. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Starting explanation

- Recall Presentation and challenge session. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain **research defense** and **next question**s using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **research defense** and **next question**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Understand the question and evidence contract

> Separate descriptive results, computational consistency and untested generalization while proposing a controlled follow-up.

Start with the research problem

A research defense can end with a limited or negative finding. Explain the original question, baseline, units, procedure, result, failures and reproduction status, then identify the gap a next study would resolve. A matching replay supports consistency of the recorded computation; it does not show that the result transfers to new people or populations.

A useful next experiment distinguishes plausible explanations instead of merely collecting easier examples.

Original concept explanation

A **research defense** distinguishes what was learned, what failed and which **next question** is justified. A complete investigation can end with a limited or **negative finding**.

Complete case inputs, rules and limits

Defense item	Complete supplied exercise contract
Main results	Candidate 12/16; baseline 11/16
Main replay	Matching replay stipulated by the scenario; not newly performed here
Main missing scope	No fresh population sample or user study
Separate picture	Frozen model: clear images 8/10, shaded images 4/10
Picture grouping	Source objects grouped within conditions; number and cross-condition pairing not supplied
Plausible alternatives	Lighting, camera or object differences may matter; cause not established
Proposed follow-up	Same held-out objects under both lighting conditions, one camera, fixed model/scoring
Changed follow-up	Camera changes as well as lighting, reintroducing a competing explanation

Name given and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace the complete main case before interpreting it

> Follow each evidence-to-conclusion step and retain unknown quantities.

Original declared procedure

Present the frozen question, credible baseline, unit design, primary analysis, uncertainty assumptions, failures and reproduction record. Propose next work that resolves a named gap with new **evidence**.

Complete main-case evidence ledger

Main defense component	Supported statement
Candidate observed rate	12/16 = 75%
Baseline observed rate	11/16 = 68.75%
Descriptive difference	One of sixteen = 6.25 percentage points
Replay matches	Supports recorded computational consistency under the supplied scenario
Fresh population / user benefit	Not studied
Significance / universal usefulness	Not supplied or established
Next work	Resolve a named evidence gap with an explicit new design, not a stronger headline

State the scope and explain the visual

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Research defense and next questions — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Study shows candidate 12/16 versus baseline 11/16; replay matches; no fresh population sample or user study.	Present the frozen question, credible baseline, unit design, primary analysis, uncertainty assumptions, failures and reproduction record. Propose next work that resolves a named gap with new evidence.	75% versus 68.75%, gain 6.25 points in 16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

Predict the changed case before reading the explanation

> Keep the rule and evidence boundary visible.

Main task and response

- Study shows candidate 12/16 versus baseline 11/16; replay matches; no fresh population sample or user study. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured change

- Change: Change cameras as well as lighting in the follow-up.

▼ Check after predicting

Prediction: The design again entangles two conditions.

Reason: A camera difference provides another explanation for any performance gap.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Connect the main result with the separate pictured case

> Keep the two scenarios and their supplied evidence distinct.

Main case and original checked reasoning

Study shows candidate 12/16 versus baseline 11/16; replay matches; no fresh population sample or user study.

75% versus 68.75%, gain 6.25 points in 16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate pictured inputs and complete explanation

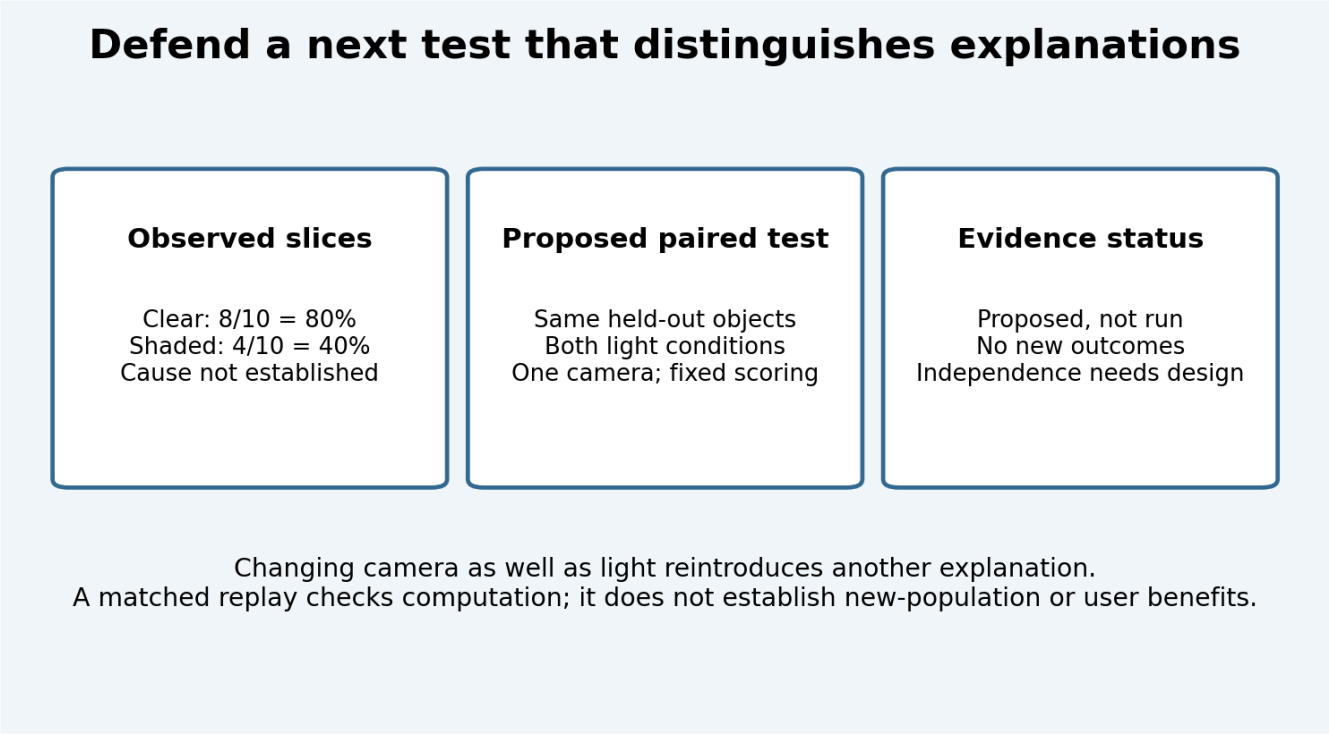
Defend the next research question. A research defence identifies an observed weakness and a testable follow-up.

A frozen model is correct on 8/10 clear images and 4/10 shaded images. Source objects are grouped within each condition.

The separate picture gives clear-image correctness eight of ten, or 80%, and shaded-image correctness four of ten, or 40%, a forty-percentage-point descriptive gap. Grouping is mentioned, but the number of source objects and whether the same objects appear in both conditions are not supplied. Lighting sensitivity is plausible; camera and object differences are also possible explanations.

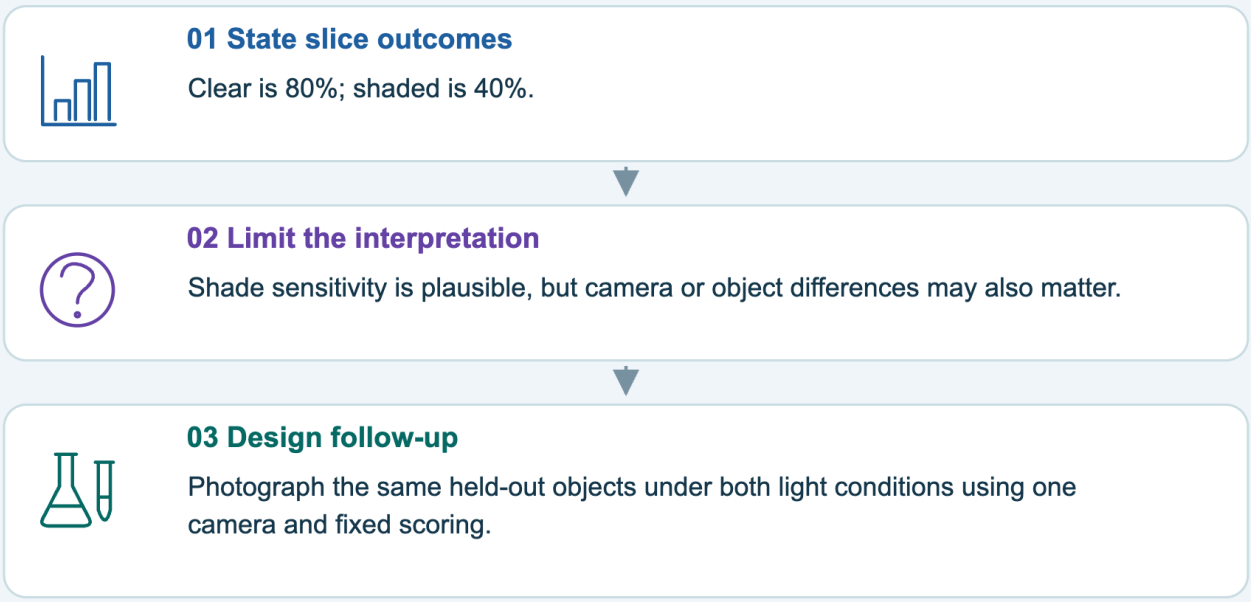
A proposed follow-up photographs the same held-out objects in both light conditions using one camera and fixed scoring. It reduces those particular differences, but still needs adequate independent objects, order/collection controls and justified analysis. No follow-up outcomes or causal finding are supplied. Changing cameras as well as lighting would again entangle explanations.

Read the labelled representation



AI ACADEMY / AI-7/36 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. State slice outcomes	Clear is 80%; shaded is 40%.
2. Limit the interpretation	Shade sensitivity is plausible, but camera or object differences may also matter.
3. Design follow-up	Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A defence should explain results, uncertainty and what would discriminate between explanations next. A controlled paired follow-up is more informative than simply collecting extra easy images.

Complete picture evidence and limits

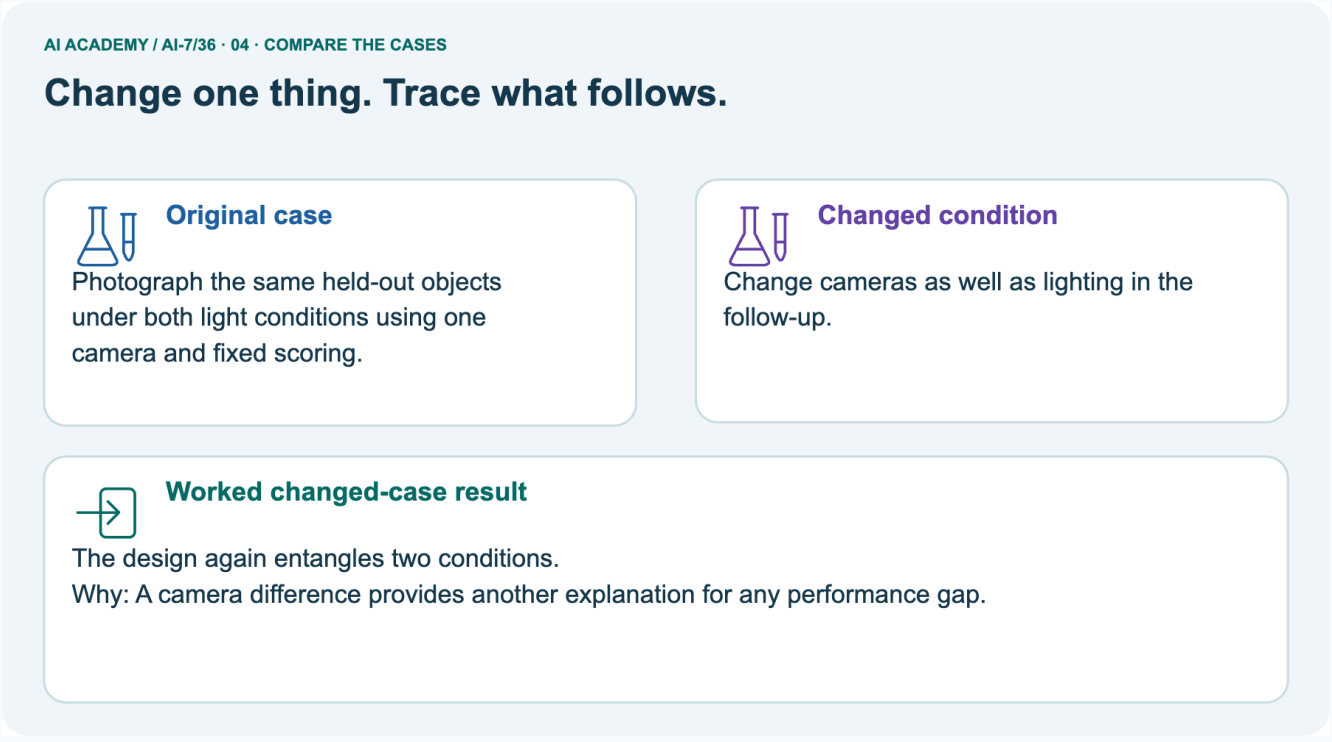
Separate defense evidence	Supported value or next-step status
Clear slice	8/10 = 80%
Shaded slice	4/10 = 40%
Observed gap	40 percentage points
Object identity/pairing details	Not supplied; no assumption of ten independent objects
Possible explanations	Light, camera or object differences; untested

Separate defense evidence	Supported value or next-step status
Proposed matched design	Same held-out objects; both light conditions; one camera; frozen procedure
If camera also changes	Alternative explanation returns
Follow-up status	Not run; no new outcome or causal conclusion

Predict the changed condition

- Change one thing: Change cameras as well as lighting in the follow-up. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what remains unknown

- Even a paired follow-up needs adequate independent objects and justified analysis; no causal conclusion is claimed yet.

- Read the labelled picture: Compute the clear and shaded rates and identify alternative explanations. How does the same-object, one-camera follow-up help, and why does changing cameras again reintroduce confounding?

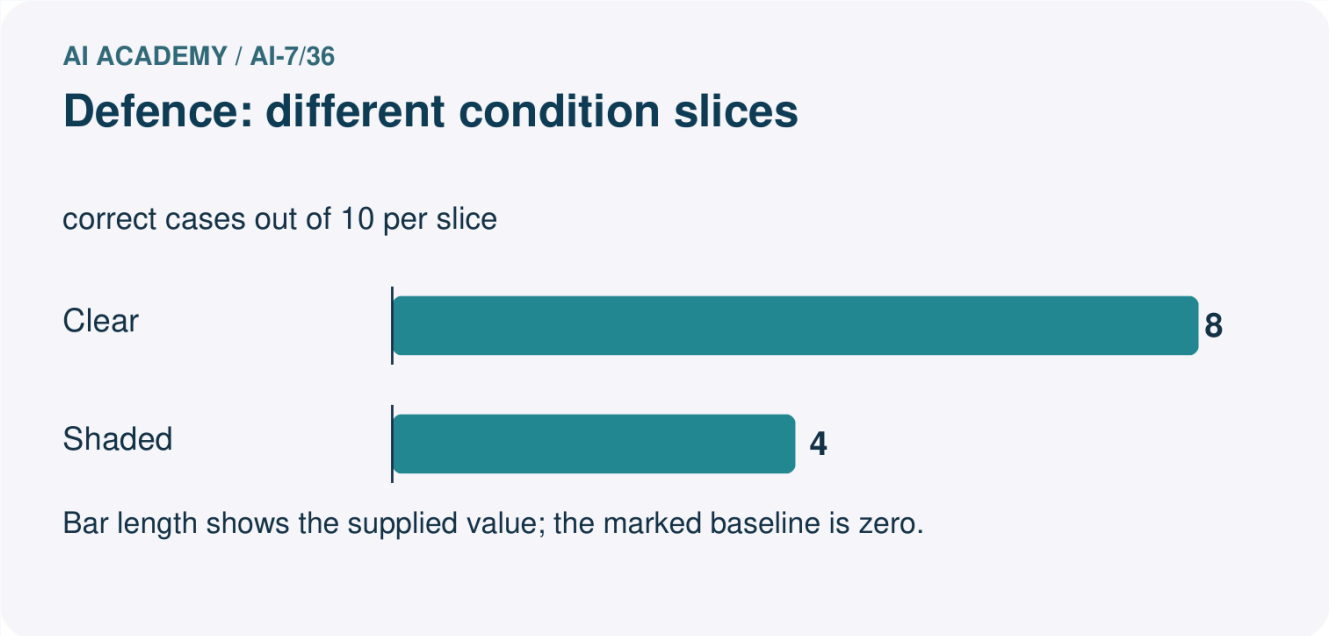
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Research defense and next questions

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/36

Research defense and next questions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State slice outcomes

Clear is 80%; shaded is 40%.



2. Limit the interpretation

Shade sensitivity is plausible, but camera or object differences may also matter.



3. Design follow-up

Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

CHANGE ONE THING

Change cameras as well as lighting in the follow-up.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Find the first unsupported inference and repair it

> Use the supplied conditions rather than inventing missing evidence.

Original qualification and repair setup

The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete boundary and misconception probes

Defense audit	Checked interpretation
Candidate one more of sixteen	Report 6.25-point descriptive gain with denominator
Replay agrees	Computational consistency, not new-data generalization
No user study	No measured user-benefit claim
Shaded slice lower	A hypothesis about lighting, not established cause
Grouped source objects	Count/independence/pairing still need actual design records
Same-object one-camera proposal	Helps distinguish explanations; not an automatic causal guarantee
Camera and lighting both vary	Confounding remains
Candidate loses in independent case	A reproducible negative result is still a valid research outcome

Write and justify the repair

- A report omits this qualification: The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **State one repair and the checks you would rerun.**

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise the mechanism with progressively less help

> Keep the selected independent case for a first attempt.

Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported workbook trace

Use workbook W2 for the connected example "**Research defense** and **next question**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for **Research defense** and **next question**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Compute the clear and shaded rates and identify alternative explanations. How does the same-object, one-camera follow-up help, and why does changing cameras again reintroduce confounding?

> **My explanation using a pictured step and the given facts**
>
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Reserve the independent task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate, record the evidence and check the limits

> Distinguish a prediction, a paper trace and an actual local execution.

Current investigation

Study shows candidate 12/16 versus baseline 11/16; replay matches; no fresh population sample or user study. Show the intermediate steps and give the conclusion with its scope.

Complete investigation instructions

1. Prepare a paper defense record with the main candidate/baseline counts, one-case descriptive difference, stipulated replay status and missing population/user evidence. Do not claim to have performed that replay. Separately compare clear and shaded slices, list alternative explanations and sketch the same-object, one-camera follow-up.

   Mark the follow-up not run, retain grouped-unit and ordering assumptions, and explain why changing cameras again weakens attribution. No model, camera collection, user study or independent reproduction is performed. Reserve the final twelve-case negative-result task for the first independent attempt.

Evidence record and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try the selected new case independently

> Report the candidate’s lower observed result and matching reproduction, then propose a specific question that would resolve an evidence gap.

Complete independent task

Project candidate 9/12, baseline 10/12; reproduction matches. State result and a useful **next question**.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Apply a new case independently

Attempt this without hints

Project candidate 9/12, baseline 10/12; reproduction matches. State result and a useful **next question**.

> **My independent procedure, result and limitation**
>
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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

</details>

MAIN LESSON · SECTION 11

Save the research artifact and unresolved conditions

> Keep supplied evidence, proposed revisions and actual work distinct.

Project task and artifact

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Project candidate 9/12, baseline 10/12; reproduction matches. State result and a useful **next question**. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Retrieve the mechanism and state the evidence boundary

> Explain what follows and what would require new evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **research defense** and **next question**s

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Research defense	A justified account of methods, findings and limits
Next question	A new bounded inquiry motivated by a gap
Negative finding	A result failing the declared expectation

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Research defense and next questions — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Study shows candidate 12/16 versus baseline 11/16; replay matches; no fresh population sample or user study.	Present the frozen question, credible baseline, unit design, primary analysis, uncertainty assumptions, failures and reproduction record. Propose next work that resolves a named gap with new evidence.	75% versus 68.75%, gain 6.25 points in 16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Defend the next research question. A research defence identifies an observed weakness and a testable follow-up.

By the end, I can	How I will show it
Explain the mechanism and terminology of research defense and next questions	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Defend the next research question. A research defence identifies an observed weakness and a testable follow-up.
- Which labelled step matches this detail? Photograph the same held-out objects under both light conditions using one camera and fixed scoring.
- 1. Design follow-up
- 2. Limit the interpretation
- 3. State slice outcomes
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Design follow-up Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

A defence should explain results, uncertainty and what would discriminate between explanations next. A controlled paired follow-up is more informative than simply collecting extra easy images.

Changed-case question and optional hint

- Changed-case practice: Change cameras as well as lighting in the follow-up. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. The design again entangles two conditions.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The design again entangles two conditions. A camera difference provides another explanation for any performance gap.

Repair and retain the record

Review Presentation and challenge session (ai-7/35). Goal: Explain the **mechanism** and terminology of presentation and challenge session. Short practice: Answer the camera-B challenge using only the twelve camera-A records.

Why is one displayed camera-B success a demonstration rather than broad **evidence** of camera-B reliability?. Return to this week and explain what you changed.

Review Independent reproduction check (ai-7/33). Goal: Explain the **mechanism** and terminology of independent reproduction check. Short practice: Locate the Q3 scoring disagreement between 5/6 and 4/6.

What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?. Return to this week and explain what you changed.

Review Writing the methods section (ai-7/30). Goal: Explain the **mechanism** and terminology of writing the methods section. Short practice: Apply the inclusive four-character rule to oak, pine and willow.

Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

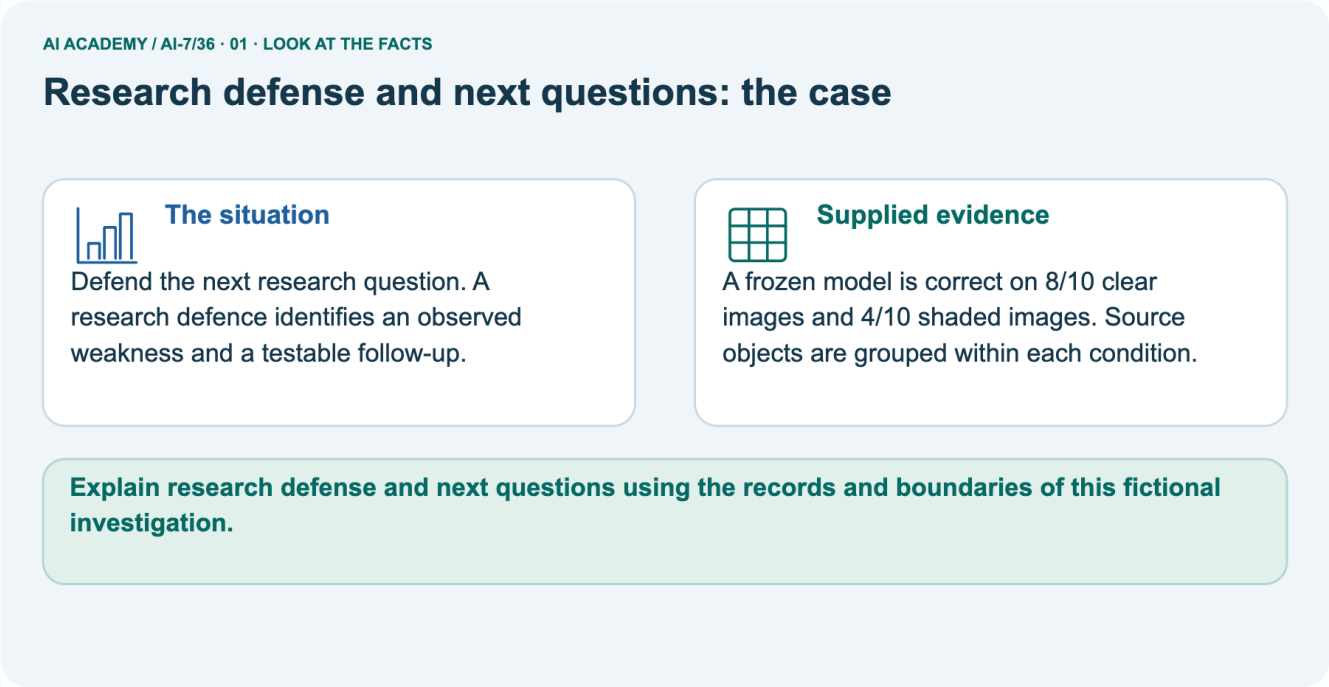
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Research defense and next questions: the case



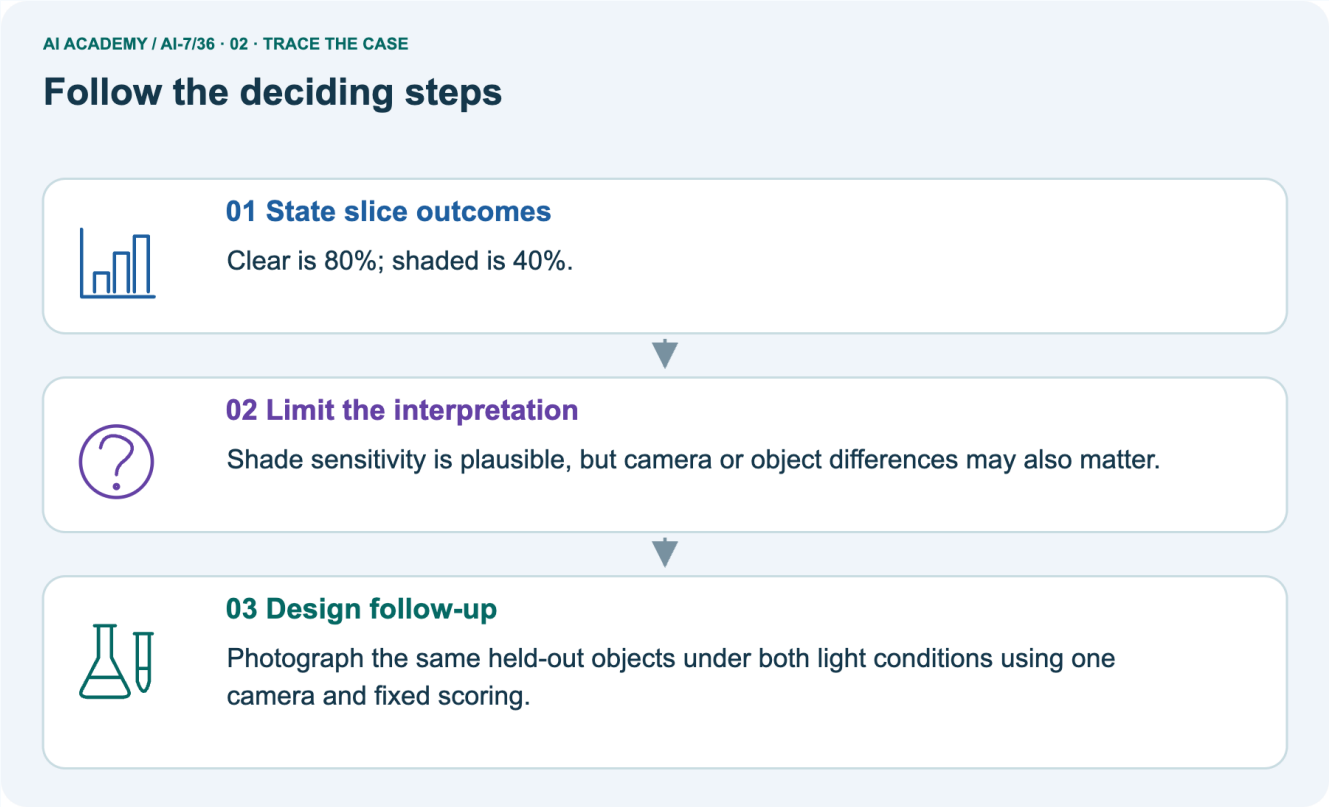
Follow the labelled picture with the matching explanation. Open the vector diagram

Defend the next research question. A research defence identifies an observed weakness and a testable follow-up. Supplied facts: A frozen model is correct on 8/10 clear images and 4/10 shaded images.

Source objects are grouped within each condition.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

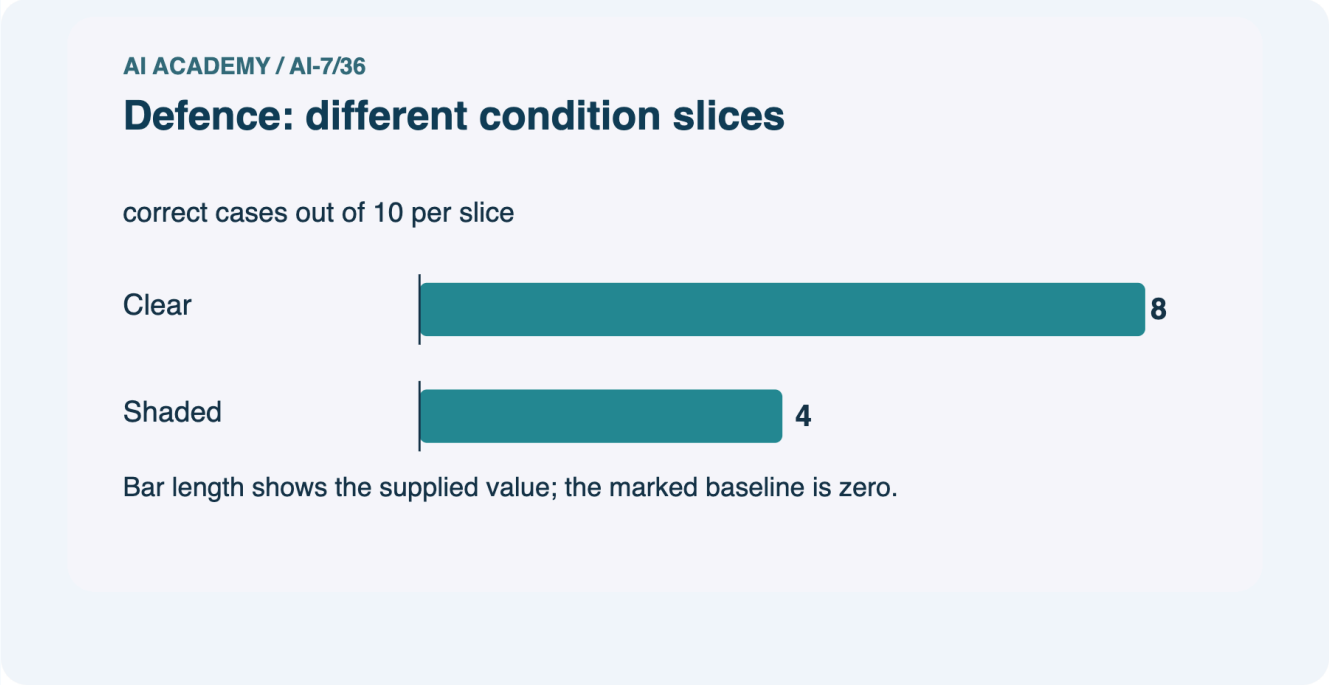


Follow the labelled picture with the matching explanation. Open the vector diagram

State slice outcomes: Clear is **80%**; shaded is 40%. → Limit the interpretation: Shade sensitivity is plausible, but camera or object differences may also matter. → Design follow-up: Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

Point to each numbered stage and explain which recorded **input** it uses.

03. Defence: different condition slices

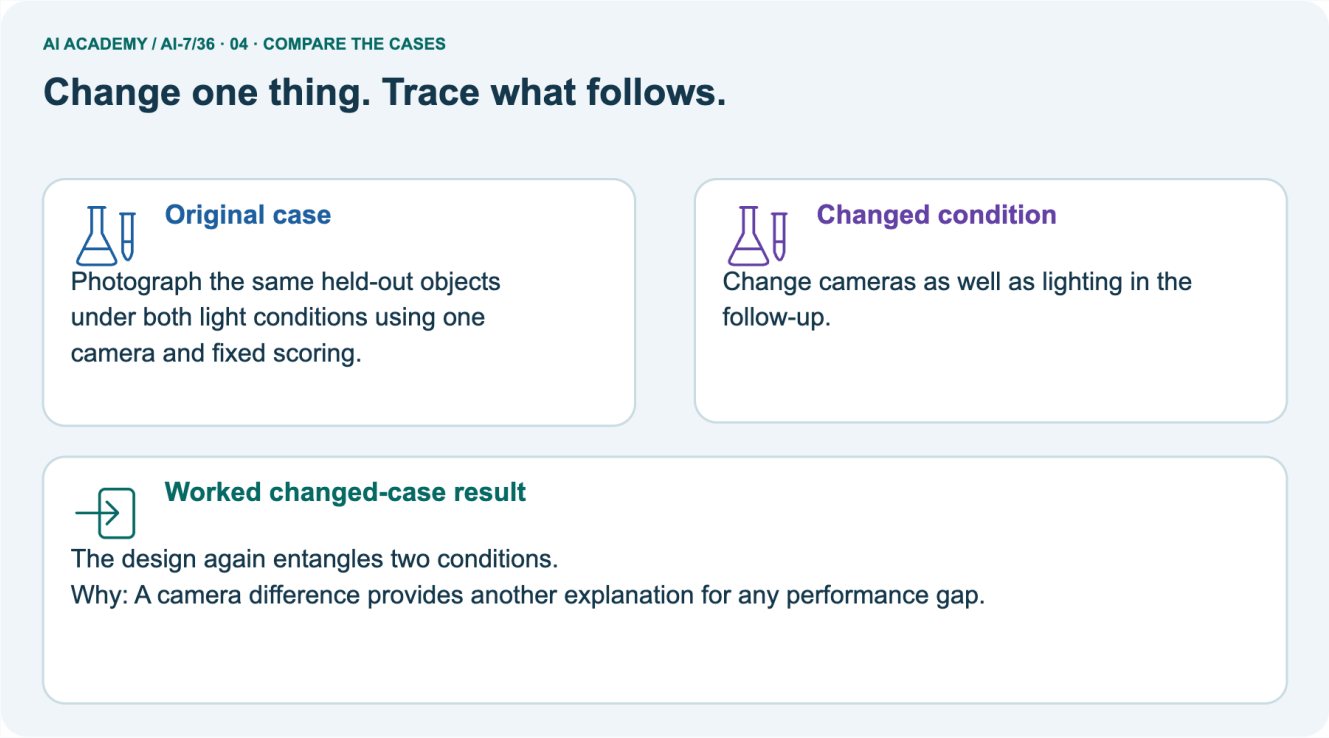


Follow the labelled picture with the matching explanation. Open the vector diagram

Defence: different condition slices. Units: correct cases out of 10 per slice. Clear: 8. Shaded: 4.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



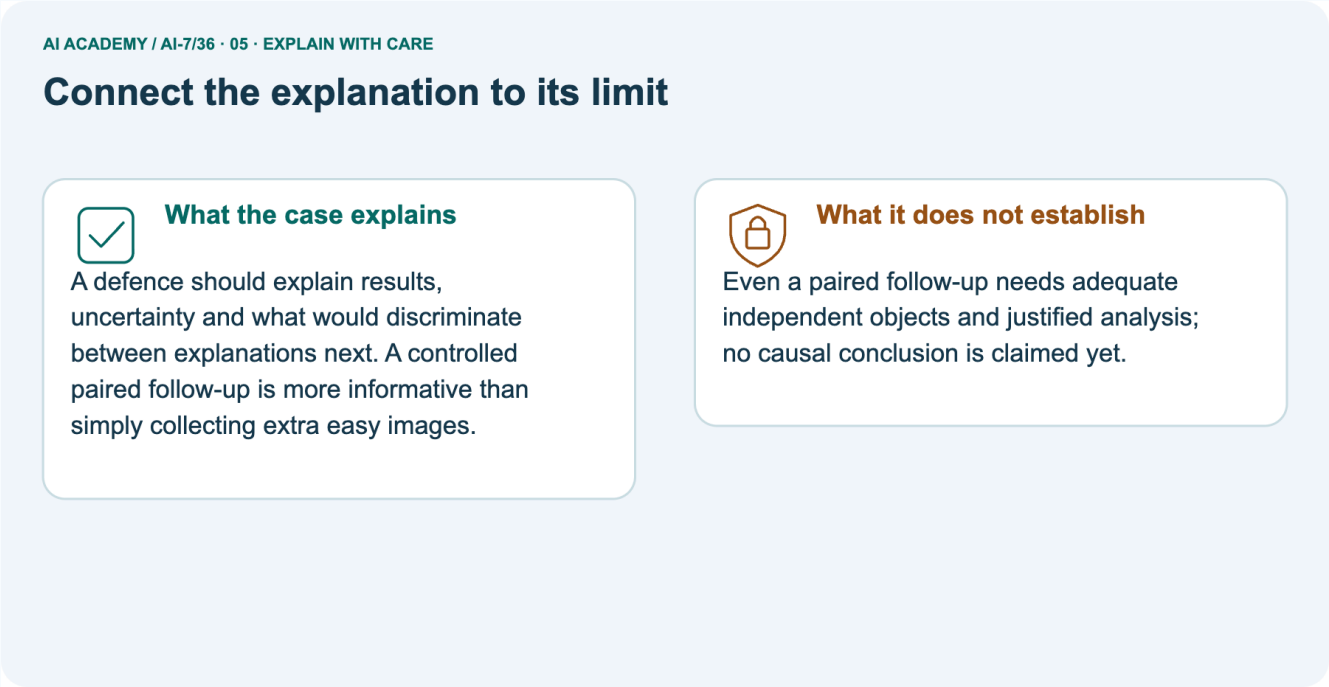
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Photograph the same held-out objects under both light conditions using one camera and fixed scoring. Changed condition: Change cameras as well as lighting in the follow-up.

Expected result: The design again entangles two conditions. Reason: A camera difference provides another explanation for any performance gap.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



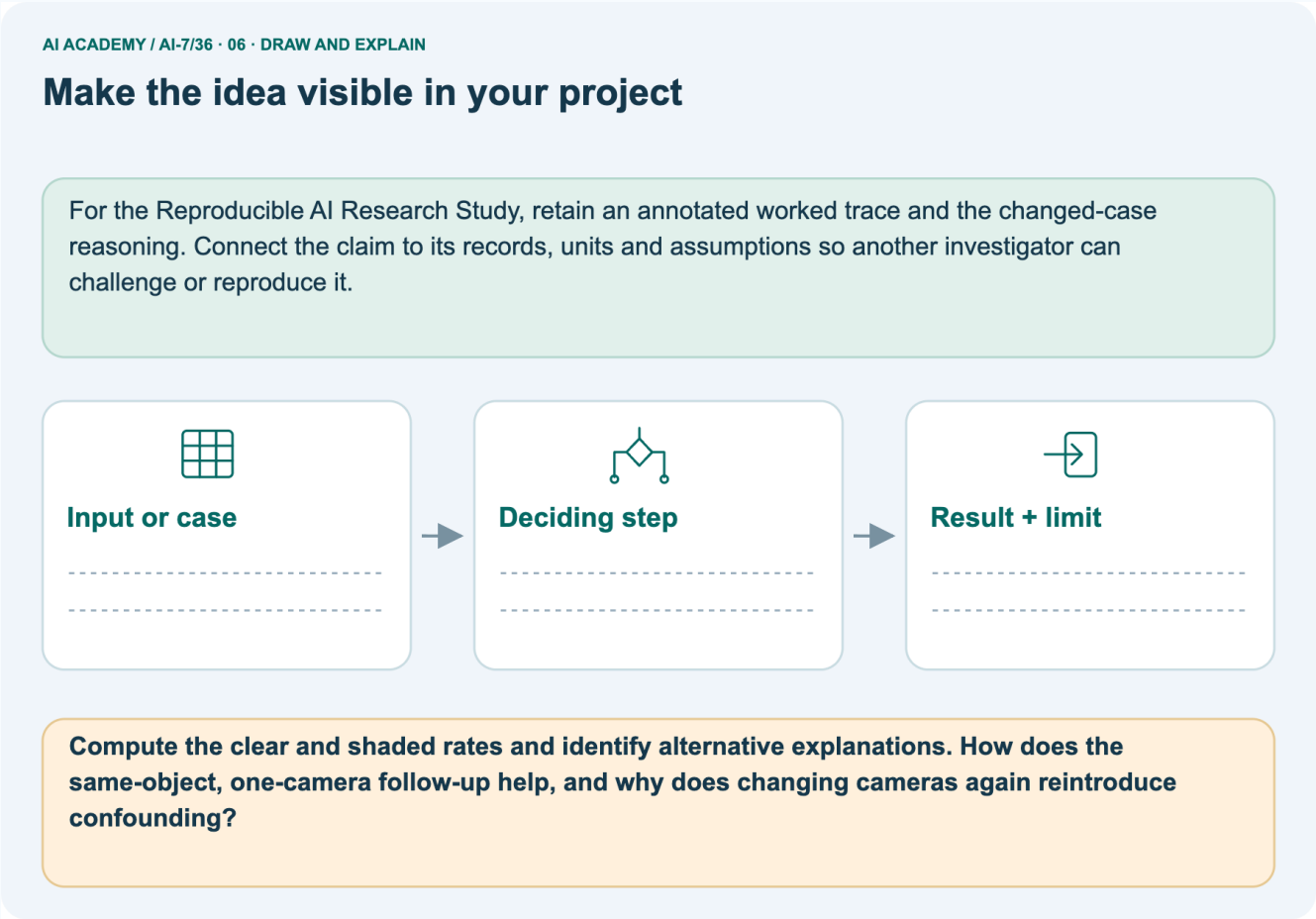
Follow the labelled picture with the matching explanation. Open the vector diagram

A defence should explain results, uncertainty and what would discriminate between explanations next. A controlled paired follow-up is more informative than simply collecting extra easy images.

Limit: Even a paired follow-up needs adequate independent objects and justified analysis; no causal conclusion is claimed yet.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Compute the clear and shaded rates and identify alternative explanations.

How does the same-object, one-camera follow-up help, and why does changing cameras again reintroduce confounding?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Research defense and next questions: the case

02. Follow the deciding steps

03. Defence: different condition slices

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.